

PROJECT TITLE : AI FAQ ASSISTANT API

REQUIREMENT ANALYSIS PHASE

The requirement analysis phase identifies the requirements needed to develop the **AI FAQ Assistant**. The system requires secure FAQ management, AI-powered content generation, search functionality, user authentication, and reliable database operations.

1. FUNCTIONAL REQUIREMENTS

The system should support user registration and login, JWT-based authentication, FAQ creation, viewing, updating and deletion, AI-powered FAQ generation using Google Gemini, keyword-based FAQ search, category management, role-based access, data validation, and centralized error handling.

2. SOFTWARE REQUIREMENTS

The project requires **Windows 10/11, macOS, or Linux, Node.js v18 or above, npm v9 or above, Express.js, MongoDB with Mongoose, Postman or Thunder Client, Visual Studio Code**, and **Google Gemini SDK** for AI-powered FAQ generation.

3. HARDWARE REQUIREMENTS

The system requires an **Intel Core i5 (8th Generation or above) / AMD Ryzen 5 or equivalent processor**, at least **8 GB RAM**, and approximately **1 GB of available storage**.

4. SECURITY REQUIREMENTS

Security is provided through **JWT authentication and authorization** and **bcrypt password hashing**. Protected routes verify authentication tokens before allowing access. Mongoose validation and centralized error handling help maintain data integrity and prevent database information leakage.

5. NON-FUNCTIONAL REQUIREMENTS

The system should be **secure, reliable, maintainable, scalable, and efficient**. It should provide standardized API responses, maintain database integrity, and use a modular architecture that supports future development.